Does the LD panel matter for PGS?

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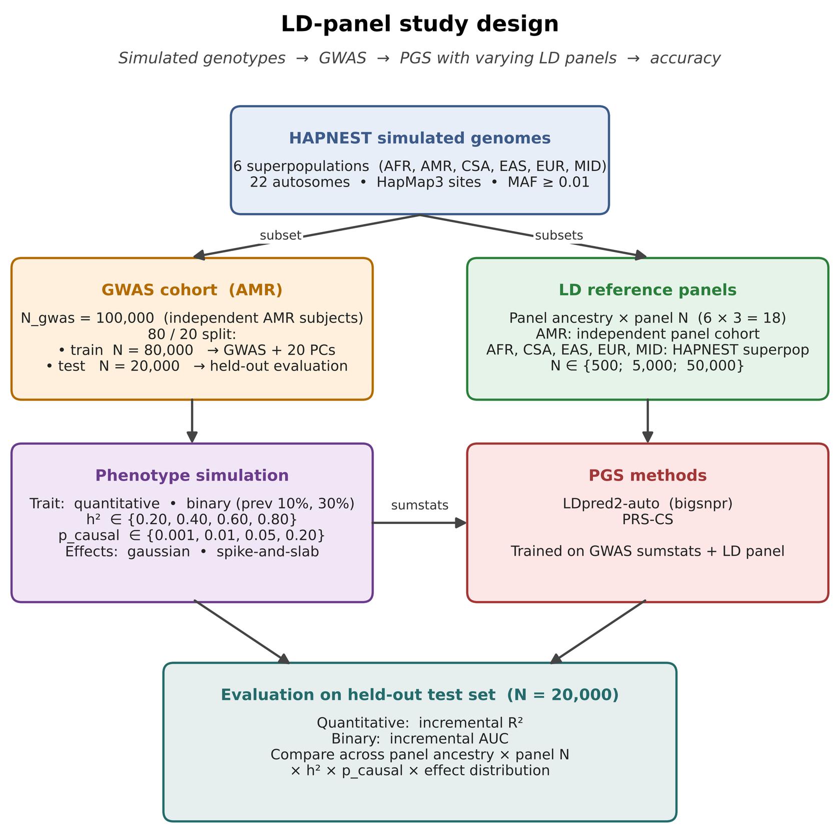
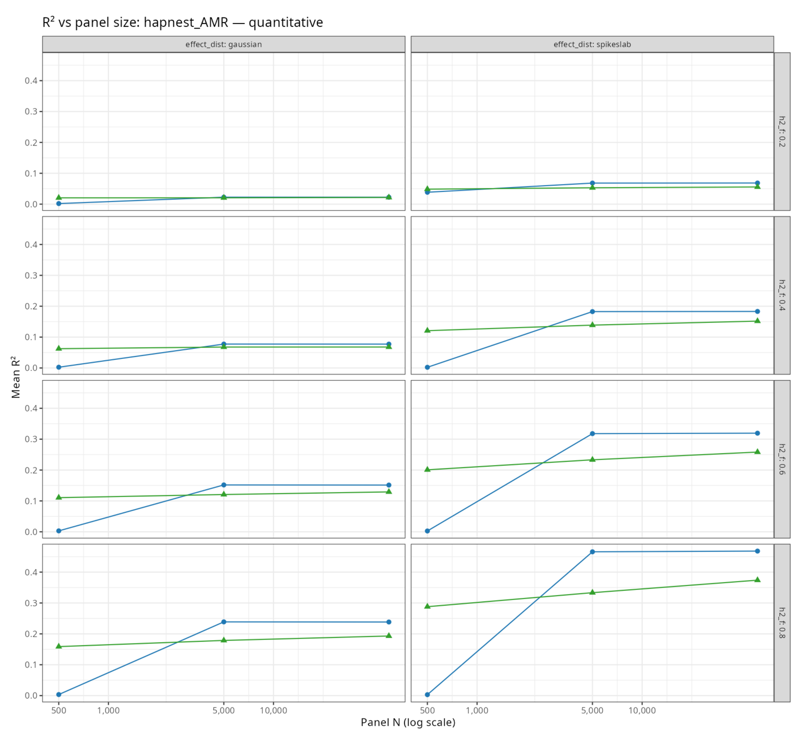
Background

Polygenic scores (PGS) trained on GWAS summary statistics need an external linkage-disequilibrium (LD) reference panel. In practice that panel is often **ancestry-mismatched** or **undersized** relative to the GWAS. How much PGS accuracy do we lose?

Simulation design



Methods compared

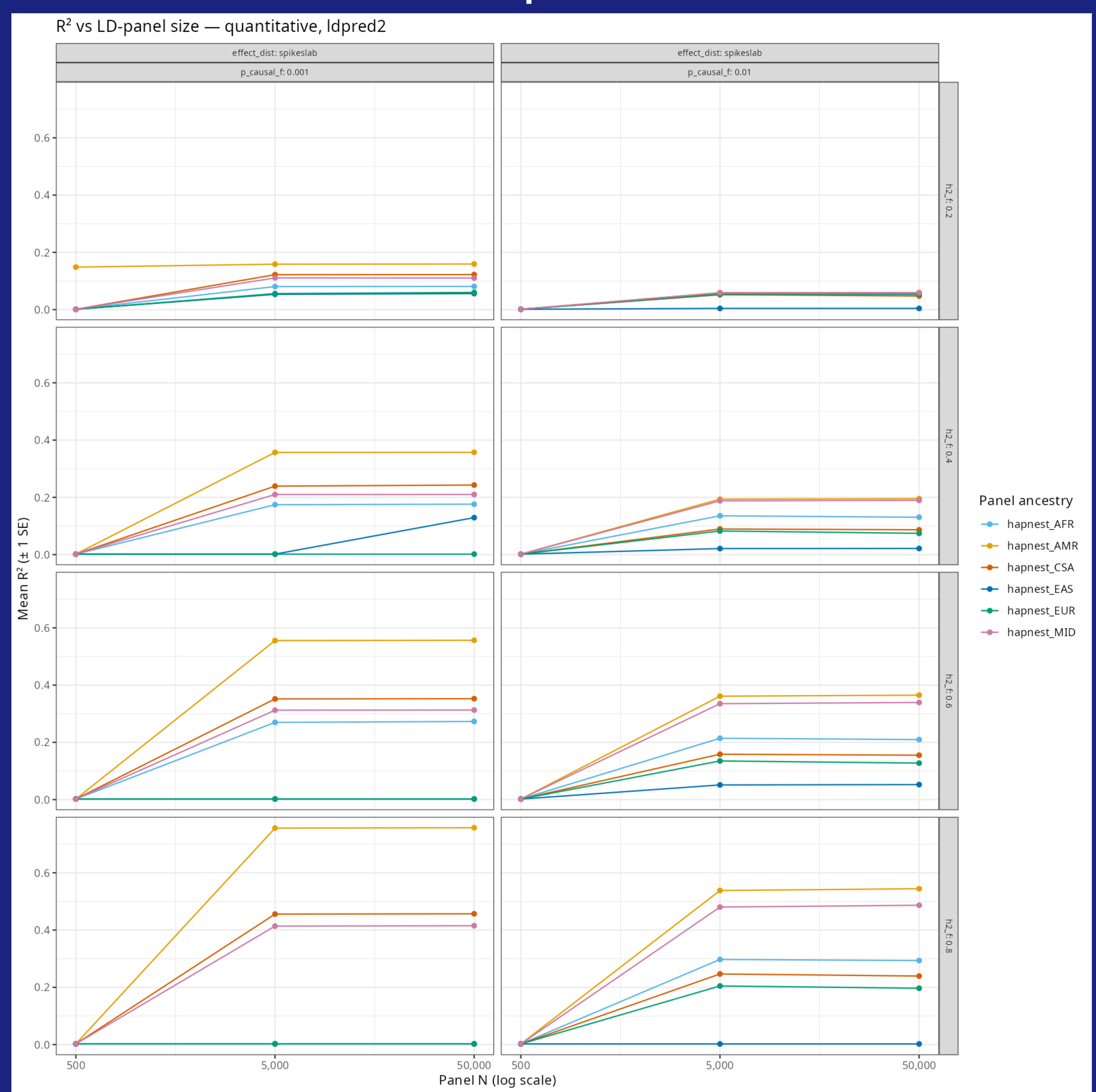


LDpred2 and **PRS-CS** produce similar results.

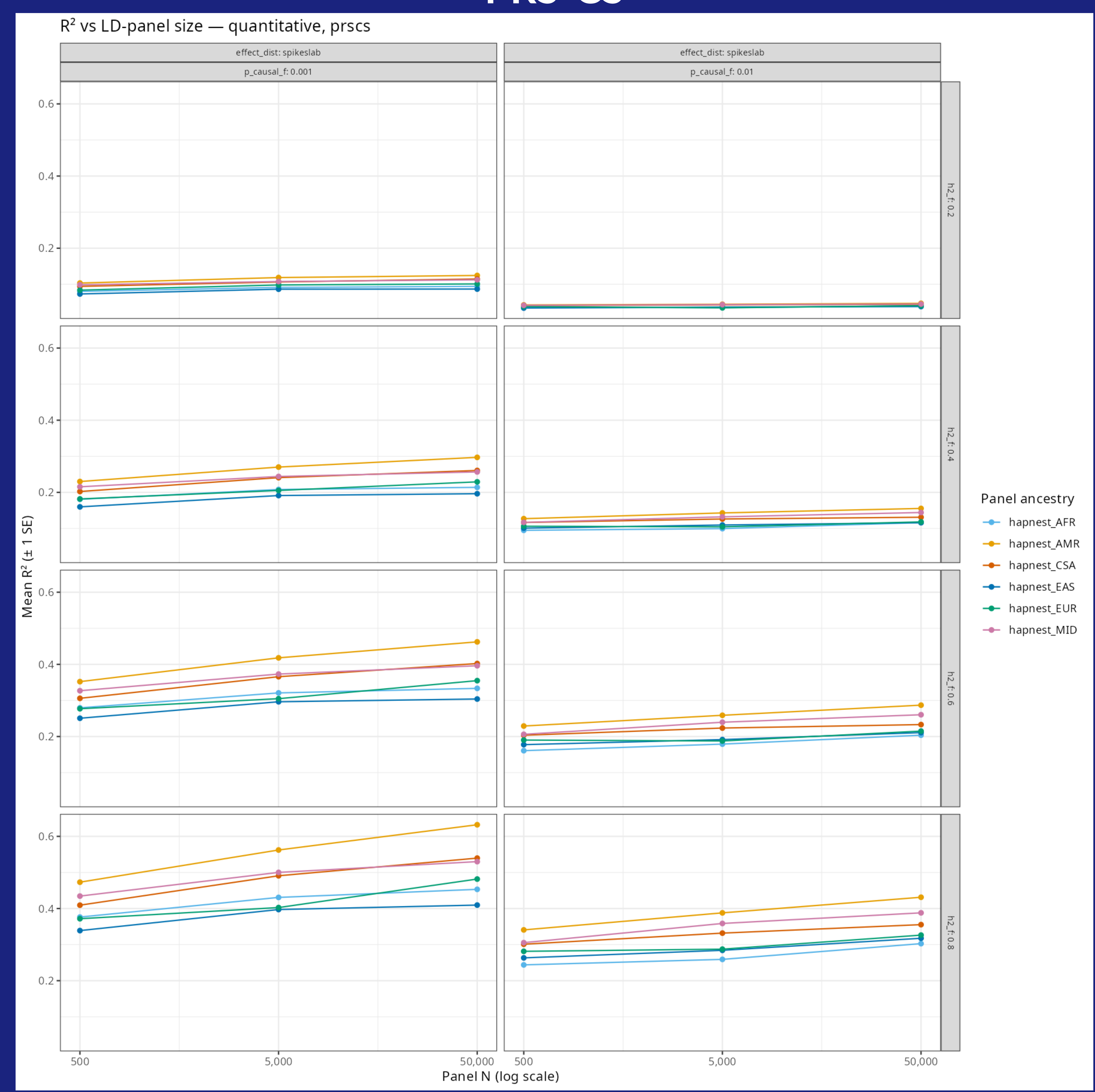
Panel ancestry and size matter when the trait is **sparse** and **highly heritable.**

In infinitesimal regimes, any panel will do.

LDpred2



PRS-CS



R^2 vs. LD-panel size, quantitative trait, sparse spike-and-slab effects ($p_{causal} \in \{0.001, 0.01\}$). Rows: heritability h^2. Matched AMR (orange) beats mismatched EAS (blue) by $\sim 2\times$ at high h^2; gap closes as panels grow.

When the panel matters

Sparse, high-h^2 traits ($p_{causal} \leq 0.01$, $h^2 \geq 0.2$): matched-ancestry AMR panels reach $\sim 2\times$ the R^2 of the worst mismatched panel (EAS).

Panel size drives most of its gain between $N = 500$ and $N = 5{,}000$; the jump to $N = 50{,}000$ adds little.

When it doesn't matter

Infinitesimal traits all six panel ancestries have simlar PGS accuracies; panel size has negligible effect.

Low-h^2 traits ($h^2 \leq 0.10$): PGS accuracy is floor-limited; panel choice is a second-order concern.

Implications

For most polygenic traits the LD-panel choice is a *minor* contributor to PGS accuracy. Investment in matched-ancestry panels pays off **primarily for sparse, high-heritability traits**.

References

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GitHub repository

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